PURPLLE STORE INTELLIGENCE

SYSTEM DEPLOYMENT & RUN GUIDE

This deployment guide provides instructions for setting up and running the enterprise-grade Purplle Store Intelligence System. The application consists of a FastAPI backend and a Streamlit analytics dashboard panel.

1. IMPORTANT: CCTV Video Feeds (Not Uploaded to GitHub)

Note on Submission Safety:

Due to GitHub file size limits and data privacy standards, raw CCTV video feeds are NOT included in the GitHub repository. To test or run the computer vision ingestion pipeline, you must manually place your video files (.mp4 or .avi formats) in the following directory:

```
data/videos/
```

2. Local Environment Setup

Run the following build command to automatically create the Python virtual environment (.venv) and install all required modules:

```
make setup
```

Once setup completes successfully, activate your environment using the appropriate script for your terminal:

On Windows (PowerShell):

```
.\.venv\Scripts\activate
```

On macOS / Linux:

```
source .venv/bin/activate
```

3. Database Ingestion & Seeding

Run this command to initialize the schemas and seed the SQLite database with premium mock stores and visitor sessions:

```
make seed
```

4. Running the System

To launch the complete application, open two separate terminal sessions and activate the virtual environment in both.

Terminal 1: Start the Backend (FastAPI)

Run the API server (available at http://localhost:8000):

```
make run
```

Terminal 2: Start the Dashboard (Streamlit)

Run the analytics dashboard panel (opens at http://localhost:8501):

```
make run-dashboard
```

5. Data Stream Simulation & Video Processing

If you want to mock CCTV stream edge vision logs directly to the active API:

```
make stream-mock
```

To run the computer vision edge-processing pipeline directly on the video files placed in data/videos/:

```
python pipeline/main.py
```